

Managing Interfaces in Wicked Problems: A Transdisciplinary Engineering Calling

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Abstract. This paper wrestles with how to manage prominent global transition challenges on local and regional scales to co-create a better world. It introduces an approach to managing navigation in wicked problems by expanding maneuverability to increase capacities to drive transformation in multi-actor decision spaces in the context of complex societal challenges. This demands a transdisciplinary engineering approach, where we propose using the concept of relational interfaces to enhance understanding across boundaries in shared problem spaces, providing a foundation for co-creation and collaboration over knowledge domains to enable transformational momentum and navigational capacity. A multiplicity of pressing issues on the global canvas urges action on local and regional scales. This paper draws on cases from regional healthcare system development, under demographic and resource-driven pressures in the Global North, and from regional energy transition planning, where diverse stakeholders require improved coordination for co-creation in the face of cultural, economic, and institutional disparities. Wicked problems often blind actors to their full scope unless addressed collaboratively. This paper emphasizes the need to surface the knowledge and experience of actors through visualizations and digital models, to foster openness and co-creation. Through enhanced understanding of how parts interact within the holistic system across silos, this is enabling actors to manage interfaces between contributions. The conclusions from the empirical cases show the potential of the identification and management of these interfaces to unlock profound change, fostering collaborative solutions to persistent societal challenges — a step toward co-creating a better world and a vital calling for future transdisciplinary engineers.

Keywords. Wicked Problems, Transdisciplinary Engineering, Transition Challenges, Co-creation, Socio-technical Systems

Introduction

Wicked problems, a term identified by Professor Horst Rittel and first documented in 1967 [1], are defined as a class of ill-formulated social system challenges, with confusing information and numerous stakeholders holding conflicting values and priorities. These problems are deeply embedded in ‘human complex systems’ [2], with interdependent elements, where both the issues and their consequences are difficult to comprehend. As the ‘shifting the burden’ archetype [3] illustrates through systems thinking, within human complex systems, the risk of addressing symptoms through misguided interventions rather than root causes is prominent when we lack understanding of the problem, potentially worsening underlying issues and creating dependency on more of the same

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fix. Within wicked problems, this phenomenon becomes intensified due to the many entanglements, where the “wicked” nature lies in the issues’ resistance to resolution, as proposed solutions often produce unintended consequences that worsen the problem [4]. This implies wicked problems resist straightforward resolution due to their evolving, multi-dimensional nature, where interventions within one part of the system can trigger unforeseen consequences across the whole. Additionally, the ‘system’ here signifies a problem within its environment characterized by an open, unknown boundary, where we aim to discern elements as investigators [5]. The role of systems thinking in this context is to support “consciously organized thinking using system ideas” [6].

As human activity is increasingly connected, it becomes increasingly hard, or even impossible, to isolate any emergence of wicked problems’ symptoms and solve their root causes. One such prominent global issue is climate change, where much hope is placed in innovations and technological disruptions to improve energy efficiency [7], while also creating energy rebound effects on global scales due to accelerating economic growth [8]. While attempts in the Global North are made to decouple the stronghold between economic growth and energy consumption, most members of the European Union are indicating an increasing gap on a domestic scale, while the figures on a global scale are unchanged due to shifting energy consumption abroad [9]. It seems that we are only playing blame games within our planetary boundaries, ultimately being ineffective and driving opposite behaviors based on a ‘techno-optimist outlook’ [10].

The complexity faced by healthcare systems can also be considered a wicked problem [11], as their issues are not restricted to one arena of play [4]. The interplay of medical, social, economic, and technological factors contributes to an ever-shifting landscape. The demographic change is considered one of the most crucial drivers of strain, where the increase in life expectancy leads to an increase in the aging population and a rising prevalence of chronic diseases. This imposes a higher demand for healthcare services with restricted resources and high-pressure working conditions, impacting employee turnover rates [12]. Beyond workforce and demographic pressures, the healthcare system is impacted by rapid technological advancements and policy constraints, and their integration is not without challenges, including ethical concerns, implementation costs, and disparities in digital literacy among both healthcare workers and patients [13].

For these reasons, global challenges such as climate change and demographic shifts demand localized and contextually aware interventions based on enhanced regional (and local) self-sufficiency. However, navigating these wicked problems, characterized by complexity, uncertainty, and stakeholder diversity, requires transcending traditional disciplinary boundaries to reshape a common future.

This paper introduces an approach that, in due course, manages coaction in wicked problems based on increased navigational capacities to co-create and drive transformation. This is achieved by processes that expand the maneuverability of multi-actor decision spaces in the context of complex societal challenges. We situate this within the emerging transdisciplinary engineering field [14], [15], which emphasizes integrating and transcending disciplinary and organizational boundaries to address complex, ill-defined, and socially relevant challenges collaboratively. In this paper, we propose the concept of ‘interfaces’ as relational and interaction-based boundaries that serve as intersections for transdisciplinary collaboration and co-creation, grounded on research in wicked problem contexts in energy optimization and healthcare systems development.

1. Theoretical Background

To navigate within the complexity inherent in wicked problems and coordinate for improved co-creation and coaction across multi-stakeholder decision spaces to drive transformation, this paper builds on five interconnected theoretical elements: 1) Defining and characterizing wicked problems; 2) Exploring interfaces as relational and interaction-based boundaries; 3) Mental models as a foundation for shared understanding; 4) Boundary objects to bridge these boundaries and elicit novel thinking; and 5) Systems thinking to enhance the management of dynamically complex problem spaces and thereby improve decision-making quality.

This theoretical foundation aligns with the perspective of transdisciplinary engineering, an emerging field that seeks to address complex, real-world challenges by integrating practical and academic knowledge, and emphasizing stakeholder co-creation, boundary-crossing, and systemic intervention in socio-technical contexts [14], [16].

1.1. Defining Wicked Problems

Wicked problems are complex societal challenges characterized by high uncertainty, multiple stakeholder perspectives, and deep interdependencies – originally introduced as a concept to describe social problems that cannot be definitively solved only managed through ongoing adaptive efforts [17]. Wicked problems has as a term been incorrectly used as a synonym for complex problems [18]. Various authors identify specific attributes that define when a problem can be considered wicked, exhibiting several attributes as described in [1], referring to [17], [18], [19]: 1) There is no conclusive formulation of a wicked problem; 2) Wicked problems are dynamic and have no stopping rule; 3) There are no true or false solutions to wicked problems, just good or bad solutions; 4) There are no immediate or conclusive solutions to a wicked problem; 5) Interventions to wicked problems provoke effects that cannot be reverted nor stopped; 6) Wicked problems do not have a definitive solution, and maybe not even a set of possible solutions; 7) Each wicked problem is unique; 8) Every wicked problem can be a symptom of another problem; 9) Different explanations can be provided to explain every wicked problem; and 10) The planner or policy maker can't be wrong as failure to address the problem has dire consequences. Wicked problems have all or several of these attributes to be considered wicked. However, there is an additional dimension to consider, which is the involvement of multiple stakeholders with different values, interests, and priorities, which adds social and political complexity [18].

Counterintuitively, searching for optimal solutions to wicked problems can be argued to be one of the drivers of how wicked phenomena arise. This is in line with the second-generation approach [17], defending that solutions develop progressively as a result of continuous discussions and critical argumentation among stakeholders, requiring a design thinking of co-evolution between solution and problem as a learning process [20]. Similarly, collaborative rationality [21], a strategy to work with wicked problems, describes the need to involve people with different views and perspectives concerned with the problem, where sharing knowledge, ideas, and divergent thinking is encouraged. This approach develops new relationships and norms of reciprocity, encourages group and individual learning about the problem, as well as double-loop learning [22] with participants redefining the problem and their goals.

1.2. Exploring Interfaces as Relational Boundaries

Interfaces generally define the interaction of two components in a modular system [23]. Introducing interfaces as relational and interaction-based boundaries in wicked problems builds on combined insights from how interfaces function in modular designs [24] and the importance of relations between system elements in systems thinking [25]. In the context of wicked problems, these two concepts may seem to contradict a suitable application. Firstly, wicked problems are not designed systems. Drawing upon the above-mentioned attributes of wicked problems, they can be presented as continuously evolving, where their importance emerges at the intersection of discovered societal issues. As human activity tries to cope with symptoms resulting from the collective emergence of intricate multi-conflict-tradeoff trajectories between actors across space and time. Secondly, standardization of interfaces is by definition impossible and non-existent in wicked problems, temptingly leading to the conclusion that it is a useless concept. Accordingly, exploring interfaces as relational herein aims first to create eyesight to see and identify potential to bridge stakeholder positions and then to be effective in addressing them.

1.3. Enhancing Understanding and Decision-Making in Wicked Problems

The theoretical elements of mental models [3], [26] boundary objects [27], [28] and systems thinking [3], [5], [6], [26] serve to expand the collective understanding of phenomena in wicked problems, i.e., a transition challenge. Their combined result can lead to an expanded navigational capacity in the problem space of wicked problems.

First, mental models are cognitive frameworks that shape how individuals perceive and interpret the world [3], [26]. However, mental models are biased, representing inner simplifications of limited perspectives. Mental models are created out of our perceptions of problems in the surrounding environment [29]. Altering mental models requires a willingness to learn [3], and double-loop learning involving questioning of underlying beliefs and governing values [22]. Additionally, shared mental models occur “when multiple individuals develop a common psychological structure for understanding their environment” [30, p. 717]. As wicked problems involve multiple actors with distinct viewpoints and transdisciplinary backgrounds, surfacing and aligning mental models becomes essential to achieving meaningful collaboration. Uncovering them can help reveal hidden assumptions, biases, and blind spots that shape stakeholder behavior and improve understanding of conflicting perspectives.

Second, the concept of boundary objects, originally conceived to abstract ill-structured solutions in transdisciplinary scientific practices [31] is adaptive to bridge across boundaries in problem-solving despite different units of analysis or methods to aggregate data, as well as foster cooperation despite different goals and demands. Later, Carlile [27], introduced a framework to facilitate knowledge integration using boundary objects to create common and shared knowledge to enhance collaboration. In the realm of novelty, i.e., transition challenges, a lack of shared knowledge often exists across stakeholder boundaries, underscoring the need for a “participatory and relational nature” of sharing and assessing new insights in this unknown landscape [27]. Finally, pragmatic boundaries are bridged through transforming knowledge using shared artifacts and methods to create the capacity to negotiate interests [27].

Third, systems thinking emphasizes consciously organizing our thinking about complex systems to improve our ability to understand change drivers and system

behavior [6]. By focusing on the dynamic complexity that emerges from relationships and interdependencies, systems thinking can utilize causal loop diagrams (CLD) and system dynamics simulations to reveal how decisions influence system states over time [3], [26]. In wicked problems, this is particularly valuable as it emphasizes optimizing system relationships rather than isolated components [25], where we cannot solve or include the complete problem, and must focus on change drivers relevant for the studied transition challenge. A system dynamics lens can help to explore how feedback loops, delays, and cross-boundary effects shape system behavior [26], where actors can better identify meaningful interventions in dynamically evolving problem spaces.

2. Learning to manage wicked problems based on real-world cases

Research activities conducted by the authors have provided an empirical ground on how to navigate prominent transition challenges on local and regional scales, such as regional development [32] and healthcare systems development [33], [34]. These cases have contributed to formulating and developing the approach presented in section 3.

2.1. Regional development case

A drastic increase in the need for green energy in an industrial region highly dependent on fossil fuels frames the case, emphasizing electrification and the policies related to driving the transformation of the local energy grids and the regional energy system. This involves a transdisciplinary multi-stakeholder situation of 15 municipalities, 21 grid owners, and industrial actors. The context fits a wicked problem as it fulfills several of the attributes defined above. Energy systems are interconnected and dynamic; a multitude of actors are involved, driven by different incentives, values, and mental models based on their respective fields of knowledge. Actors, such as grid owners, municipalities, and industries, are part of a larger system. Moreover, the transition of energy systems invokes many conflicts of interest, not only between stakeholder groups but also among citizens, when invisible greenhouse gas emissions need to be replaced by visible infrastructure.

The objective has been to increase an understanding of the participating actors on how to address the emergent transition needed to radically increase electrical power generation and to establish a shared plan across municipalities for energy planning, enabling electrification [32]. One of the biggest challenges was the perplexing complexity and the difficulty of discussing common solutions, resulting in modest activities for green transition in the stakeholder arena.

During the study, the issues associated with the transition of the energy system have been identified by involving various relevant actors. This created an understanding of different relational components of the energy system, which, if better managed, would facilitate coaction, shaping the understanding of potential interfaces among system components. As a response, visualizations of roles and responsibilities were developed, a governance structure was put in place, and system dynamics simulations of the energy system were developed to discuss potential scenarios [32]. These represent boundary objects for communication, from simple visualizations to more technically advanced.

Further efforts resulted in establishing guidelines for local comprehensive planning, proposing and enabling dialogue between the municipalities, grid owners, and industries, where new venues for iterative co-creation through new shared knowledge emerged.

Results include a political agreement at the regional level, adapted through recurring dialogues with policymakers, who also adopted the role of meta-governors.

This effort has led to important learnings, such as the power of simple visualizations as effective boundary objects to create improved interfaces by addressing relational boundaries, leading to the emancipation of shared knowledge and understanding.

2.2. Healthcare systems development case

The region of West Sweden works actively on reshaping its definition of the healthcare system for the future, by initiating simulation-based studies for closer care strategies for elderly patients [33] and for developing strategic transitions toward 2040 [34]. Current identified pressing issues are, amongst others, scarcity of resources, increasing demand for healthcare services, and technological change. The definition of how healthcare systems should be designed and planned is considered a wicked problem by embodying all or several of the attributes described above. Firstly, there is no single, conclusive problem definition of the pressing issues. Some stakeholders perceive it as a resource allocation issue, others as a technological challenge, while others focus on equity, access, or quality of care. Healthcare is inherently dynamic and has no stopping rule, as medical advancements, demographic shifts, and policy changes continuously reshape the landscape. There are no true or false solutions, as different approaches yield varying trade-offs depending on ethical, economic, and political priorities. Moreover, interventions in healthcare may provoke irreversible effects, such as changes in workforce structures or financial burdens impacting irreversible patient outcomes. The healthcare system's complexity is further exacerbated by interdependencies with economic stability, technological progress, and societal well-being, making it both a unique challenge and a symptom of broader socioeconomic issues. Additionally, multiple stakeholders with conflicting values and priorities, including policymakers, healthcare providers, patients, insurers, and technology developers, contribute to the system's social and political complexity, often leading to tensions over decision-making and resource distribution. Given these characteristics, healthcare systems cannot be solved definitively but must instead be continuously managed through adaptive, multi-stakeholder approaches that acknowledge their evolving and systemic nature.

Despite learnings about the difficulty of formulating the problem to be addressed in a previous simulation study [33], working with a completely new group revealed new layers of challenges. The objective of the latter study was to support policymakers in testing different strategies to transition towards a desired future healthcare system. One of the first steps was to interview 42 stakeholders in various positions, such as politicians, hospital directors, primary care directors, heads of emergency departments, heads of medicine and drug readiness, heads of digitalization, etc. Questions were asked to surface their mental models, i.e., their understanding of what this transition meant, its challenges, and enablers. A low level of shared understanding and a low overlap of stakeholders' mental models, often restricted by the scope of their daily work and position, was found. The interview results were shared as an interface to provide preconditions for future work, providing ground for shared understanding. During the interviews, as the understanding of the holistic system perspective evolved, a conceptual model including feedback reasoning using CLD with familiar objects to them was created. This boundary object was used in several levels of dialogue to create awareness of the type of systems thinking perspective a future system dynamics model could have. As a result of multiple interventions, specific areas of the problem were finally selected for a deeper analysis

using a system dynamics model. This model was used as a boundary object to cultivate learning about how this simulation approach could be relevant to the stakeholders' needs in future strategic work. By connecting dimensions normally studied in isolation, the simulation model could explore desired and undesired consequences of taking different actions. These interfaces and boundary objects served as a basis for productive discussions, leading to stakeholders' expanded mental models of the potential of these types of assets in their strategic development.

Important learnings from these cases include the difficulty in defining the problem to be addressed, which is very common according to the attributes that characterize a wicked problem, the importance of stakeholder engagement in the discussions about the studied problem, and the usefulness of boundary objects for quality discussions.

3. Proposed Approach to Manage Wicked Problems

This paper rests on the basis that addressing wicked problems begins by searching for the next insight across boundaries, considering various viewpoints in a more open collective system boundary that meets community goals [31], to breach the current level of shared understanding and create conditions for desired transitions. Consequently, we need to search for approaches to foster learning journeys. The aim is not to find optimal solutions, but to increase shared understanding of how the problem can evolve in a systematic procedure, by eliciting and progressing stakeholders' mental models, increasing coordinated coaction and stakeholder engagement, resulting in better decision quality that will support the management of wicked problems.

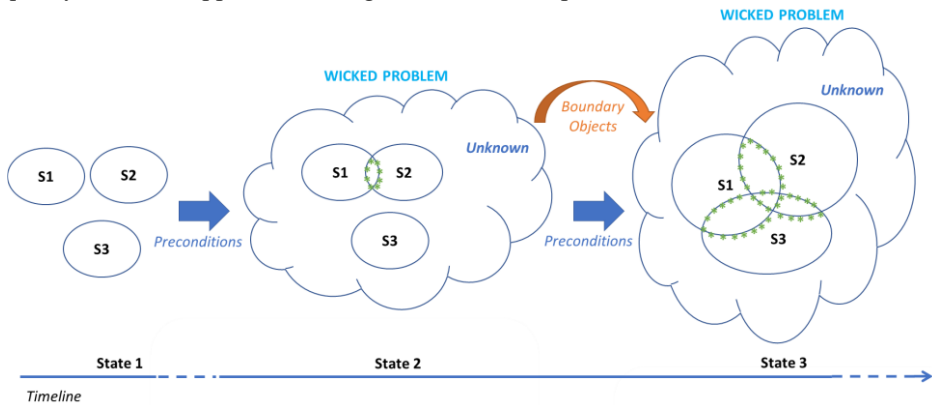


Figure 1. Illustration to support articulating the approach to manage wicked problems.

The proposed approach, illustrated in Figure 1, is based on working with wicked problems, as presented in Section 2. The theoretical standpoint is based on systems thinking and highlights the importance of using relational interfaces and boundary objects to enhance shared understanding. The use of collaborative rationality as a precondition to start any collective analysis and to perceive shared problem spaces.

Each of the circles represents different stakeholders (S), and the size of the circles represents their understanding of the problem based on aspects such as perspective, knowledge, mental models, etc. The clouds represent the dynamic problem space (that changes shape with time, actions taken, or events), characterizing the wicked problem.

The intersections between circles represent relational boundaries between stakeholders that can be discovered and created through interfaces (represented as “*” in the figure), which, as they appear, create overlap and represent enhanced maneuverability for co-action. The timeline at the bottom of the figure illustrates the evolution through different states, where State 3 depicts expanded and shared understanding compared to States 1 and 2, indicating increased capability to navigate the wicked problem.

State 1 refers to the initial status, involving immature capabilities to co-create due to low awareness, as stakeholders have their mental model and a siloed understanding of the transition challenges they face. In State 1, there is low stakeholder interaction, characterized by positioning and intentional discussions without coaction. Moving from State 1 to State 2 requires some preconditions to be met. These include the willingness to work according to collaborative rationality [21], as well as ensuring different views and perspectives are represented, i.e., a transdisciplinary stakeholder group.

State 2 is characterized by discovered and actively managed interfaces that intersect on identified commonalities across system boundaries to emancipate transition dynamics to evolve (by removing limiting conditions and nurturing drivers [3]). As stakeholder understanding of the underlying conditions grows, as problem conditions change, or when the capability of the stakeholders evolves when grasping more and more of the aspects of the wicked problem, these interfaces should evolve too. Stakeholders start building a shared understanding and expanding their mental models towards building a broader systemic view. Visualization techniques, information sharing, and creative discussions will serve as powerful interfaces in this state. As a result of the learning processes, stakeholders are becoming increasingly aware of their limited knowledge about the wicked problem, and as they uncover their unknown unawareness [35] they grow capabilities to identify and navigate the complexities and uncertainties that the unknown space represents in the problem space.

To evolve to *State 3*, the same preconditions are required, however, the role of boundary objects to create an enhanced and shared understanding and foster iterative co-creation cycles to allow for dynamic problem-solving will be central. These boundary objects may range from simple representations to more advanced system models and simulations to enhance system performance across stakeholder boundaries, to evaluate transformations, and to be able to adapt approaches to manage wicked problems. Boundary objects' crucial role in improving interfaces provides tangible means for actors to bridge gaps in language, knowledge, and priorities. The changed circles represent the stakeholders' evolution in understanding the problem at hand, through time. This shared understanding establishes ground for quality decision-making, allowing for improved co-creation and coaction. State 3 is far from being the last state; the discontinuous line in the timeline represents a process evolving as the stakeholders continue to grow in their shared understanding and navigational capacities to manage their wicked problems.

4. Discussion and Conclusions

The proposed approach to managing wicked problems is all about creating preconditions and systematic approaches to enable shifting perspectives and emancipate coaction. The presented approach does not provide a definitive solution to wicked problems, as they can't be solved, but rather a guiding approach for achieving enhanced co-navigation capacities. We acknowledge the inherent limitations of predicting outcomes in wicked

problem contexts. By introducing relational interfaces, the proposed approach focuses on searching for commonalities across a better-formed or searched-for community goal. At the core, this is believed to facilitate creating the required foundations at the stakeholder positions, to enable building bridges of understanding supported by boundary objects. Our aim is not to control or standardize the interfaces, but to use them to facilitate the redefinition of system boundaries as we co-create the higher-level goals across stakeholders rather than being stuck in localized and narrow solutions to global transition challenges. In doing so, the paper contributes to operationalizing transdisciplinary engineering principles by demonstrating a practical approach to managing interfaces in wicked problems.

Wicked problems, by their very nature, resist definitive solutions and demand continuous adaptation and collaborative engagement. This paper has introduced an approach to managing wicked problems by proposing how enhancing maneuverability within multi-actor decision spaces can be achieved by proposing relational interfaces to discover relational boundaries and to facilitate creating boundary objects that more precisely bridge knowledge gaps and disparity among stakeholders. By recognizing that wicked problems require ongoing navigation rather than resolution, we emphasize the importance of fostering shared mental models, facilitating relational interactions, and employing systems thinking to expand collective and increasingly shared understanding. The proposed approach, informed by real-world applications in regional energy transition and healthcare system development, highlights the value of iterative learning and co-creation processes in addressing complex societal challenges.

Future research should further explore how the introduced approach can be refined and adapted to different contexts, ensuring that the management of wicked problems remains a dynamic and evolving endeavor. Transdisciplinary engineers are uniquely positioned to lead such transformation endeavors, where managing interfaces becomes a central capability for fostering societal resilience and innovation. This underscores the need for educational and institutional environments that support the development of these competencies — a calling for the future of transdisciplinary engineering.

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